Manvi

+91 7060099187 manvii1503@gmail.com linkedin.com/in/manvyy

Professional Summary

Results-driven Computer Science undergraduate with a strong foundation in machine learning, full-stack web development, and data engineering. Proficient in Python, Java, and the MERN stack, with hands-on experience building NLP-based predictive models and scalable REST API-driven applications. Certified in Google Data Analysis and Oracle AI Foundations, seeking to leverage technical expertise to build innovative, data-centric software solutions.

Education

Kalinga Institute of Industrial Technology July 2023 - May 2027
Bachelor of Technology in Computer Science & Engineering *Bhubaneswar, Odisha*

- **Relevant Coursework:** Introduction to C Programming, Data Structures and Algorithms (C), Object Oriented Programming (Java), Database Management System (Oracle Database Express Edition XE), Computer Network, Design and Analysis of Algorithms, Artificial Intelligence (AI), Machine Learning (ML)
- **Relevant Elective:** Creativity, Innovation, and Entrepreneurship, Cloud Infrastructure Management Services with AWS, Data Mining & Data Warehousing, Distributed Operating System

St. Aerjay Public School April 2021 - April 2022
Intermediate in Non-Medical with Computer Science *Bulandshahr, Uttar Pradesh*

- **Relevant Coursework:** Introduction to Python Programming, Implementing Basic Data Structures in Python, Database Management System (MySQL), Connecting MySQL to Python

Projects

Sentiment Analysis for Movie Reviews | *Python, Scikit-learn, NLTK, Pandas, IMDB Dataset*

- Developed and trained a supervised machine learning model to classify 50,000+ IMDB movie reviews as positive or negative, achieving a high accuracy rate on the test set.
- Engineered a robust text preprocessing pipeline using NLTK to handle noise reduction, implementing tokenization, lemmatization, stop-word removal, and regex-based cleaning.
- Transformed textual data into numerical feature vectors using TF-IDF (Term Frequency-Inverse Document Frequency) to capture word importance and improve model interpretability.
- Evaluated multiple classification algorithms, including Logistic Regression, Naive Bayes, and Support Vector Machines (SVM), performing hyperparameter tuning to optimize performance.
- Visualized model performance using Matplotlib/Seaborn to plot confusion matrices and ROC curves, ensuring balanced precision and recall scores.

Sentiment Analysis of Amazon Review Dataset | *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

- Designed a sentiment analysis system to process and analyze a large-scale dataset of Amazon product reviews, extracting actionable insights regarding customer satisfaction.
- Implemented Exploratory Data Analysis (EDA) to identify trends in customer feedback, visualizing the distribution of ratings and review lengths to understand data characteristics.
- Built a predictive model to correlate textual review content with star ratings, utilizing Natural Language Processing (NLP) techniques to handle unstructured text data.
- Addressed class imbalance in the dataset by applying resampling techniques, resulting in a more unbiased model capable of detecting negative sentiment effectively.
- Generated comprehensive reports highlighting key sentiment drivers, demonstrating the model's potential application in improving product quality and customer service strategies.

CampusCare Complaint Management System | *HTML, CSS, PHP, MySQL, REST API*

- Developed a multi-tier student complaint management system categorizing issues for Mentors (academic/general), Wardens (hostel), and IROs, with an escalation path to Admin review.
- Engineered a secure backend architecture using PHP and REST API principles, implementing JSON Web Tokens (JWT) for authentication and Role-Based Access Control (RBAC).
- Designed a user interface with HTML and CSS allowing students to submit complaints, check real-time status updates, and trigger escalations if unsatisfied with the resolution.

- Architected a MySQL relational database to manage complaint lifecycles, tracking statuses from Pending and In Progress to Resolved or Rejected.
- Built dedicated role modules enabling authorities to fetch assigned complaints, take action, and add explanatory remarks while ensuring strict separation of concerns.

Single Image Super Resolution using CNN & GAN | *Python, PyTorch, OpenCV, Deep Learning*

- Developed a deep learning pipeline utilizing Convolutional Neural Networks (CNN) and Generative Adversarial Networks (GAN) to upscale low-resolution images by 4x while preserving high-frequency textures.
- Implemented and trained an SRGAN (Super-Resolution GAN) architecture, optimizing both adversarial and perceptual loss functions to generate photorealistic high-resolution outputs.
- Processed and augmented the large-scale Flickr2K dataset, comprising over 2,600 high-resolution images, using OpenCV and NumPy to construct robust paired training samples.
- Evaluated model performance using quantitative metrics including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM), outperforming baseline interpolation methods.
- Engineered an efficient inference script for rapid batch processing, significantly enhancing image visual quality for downstream computer vision applications.

Technical Skills

- **Languages**: HTML, CSS, PHP, JavaScript, C, Python, Java, SQL
- **Frameworks & Libraries**: React, Express, Node.js
- **Machine Learning & AI**: TensorFlow, Scikit-learn, NLTK, Pandas, NumPy, Matplotlib, Seaborn
- **Cloud Computing**: AWS, Cloud Services, Cloud Infrastructure Management
- **Databases**: MongoDB, MySQL, Oracle SQL
- **Tools & Platforms**: AutoCAD, Kaggle
- **Core Competencies**: Web Development, Data Structures, Algorithm Design, Operating Systems, DBMS, Computer Architecture, Computer Networking, Distributed Systems, Software Design, Software Architecture, Data Mining, Data Warehousing

Certifications & Licenses

Cloud Security Engineer - Google Cloud Certification — July 2026
Coursera (Google Cloud Training) — *Credit included for KIIT*

Google Data Analysis with Python Specialization — Mar 2026
Coursera (Google)

OCI AI Foundations Associate — Sep 2025 - Sep 2027
Oracle Cloud Infrastructure

Data Engineering — April 2026
ExcelR — *Credit included for KIIT*